

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
II Sem B.Tech (CBCS) Mid Semester Examinations - II, April 2023

PH1001-1 – ENGINEERING PHYSICS

Duration: 1 Hour

Max. Marks: 20

*Note: Answer any **One** full question from **each Unit**.*

List of constants: Velocity of light, $c=3 \times 10^8 \text{ ms}^{-1}$, Planck's constant, $h=6.63 \times 10^{-34} \text{ Js}$,
 Electron mass, $m=9.11 \times 10^{-31} \text{ kg}$, Electron charge, $e=1.6 \times 10^{-19} \text{ C}$,
 Boltzmann constant, $k=1.38 \times 10^{-23} \text{ J/K}$, Avogadro number, $N_A = 6.023 \times 10^{26} / \text{kg mole}$.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Explain the terms a) Fermi Energy, b) Mobility and c) n- type semiconductors	3	L*1	3	1,2
b)	Obtain an expression for the conductivity of an extrinsic semiconductor.	4	L2	3	1,2
c)	The critical temperature and critical magnetic field for superconducting lead are 7.2 K and 800 Gauss respectively. What will be the temperature up to which lead will be in superconducting state in a magnetic field of 400 Gauss?	3	L3	4	1,2
2. a)	Explain the effect of temperature on the conductivity of an intrinsic semiconductor.	3	L1	3	1,2
b)	Explain the properties of superconductors with suitable diagram.	4	L2	4	1,2
c)	A semiconductor sample of thickness $1.8 \times 10^{-4} \text{ m}$ is placed in a magnetic field of 0.5T acting perpendicular to its thickness. Find the Hall voltage generated when a current of 200 mA passes through it. Assume the carrier concentration to be 10^{23} m^{-3} .	3	L3	3	1,2
Unit – II					
3. a)	Write any three difference between Direct and Indirect band gap semiconductors.	3	L1	3	1,2
b)	What is Hall effect? Derive the expression for carrier concentration and hall coefficient in terms of mobility of the charge carries.	4	L2	3	1,2
c)	The compound gallium arsenide has an intrinsic conductivity of $10^{-6} \text{ ohm}^{-1} \text{ m}^{-1}$ at 27 °C. How many electrons have jumped the forbidden energy gap? [Given: $\mu_e = 0.88 \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$ and $\mu_h = 0.04 \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$].	3	L3	3	1,2
4. a)	Write any three difference between Type I and Type II superconductors with suitable diagrams.	3	L1	4	1,2
b)	Explain Fermi factor. Discuss probability of occupancy of electrons at $T = 0 \text{ K}$, and $T > 0 \text{ K}$.	4	L2	3	1,2
c)	The electrical conductivity of an intrinsic semiconductor increases from $19.96 \text{ ohm}^{-1} \text{ m}^{-1}$ to $79.44 \text{ ohm}^{-1} \text{ m}^{-1}$ when the temperature is increased from 60 °C to 100 °C. Find the band gap energy of the semiconductor.	3	L3	3	1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
